D 153 | Equal ages, separate meanings

PRIVATE FICTIONAL D 2 C practice. Rupees; monthly acquisition cohorts T 1-T 14, as of end T 14. No real brand/customer records, accounting opinion, analytics connection or spend.

Cohort definition	Each cohort starts with 1000 distinct retained first-order buyers. Age 0 acquisition; age a for cohort c occurs calendar T(c+a). Repeat counts are distinct monthly buyers, one retained paid order each at that age. A person may appear in several ages; sum is order events, not distinct-ever-repeat people.
Eligibility	Fourteen cohort rows/ages 0-6 only. Future cells show N/A, not zero. Ages 7+ not supplied even if some old cohorts have reached them. At endT 14 ages 1/3/6 eligible 13/11/8. T 10 age 6 is unknown. As-of edits restricted 1-14, not a forecast.
Patterns	T 1-6 repeat 250 / 180 / 150 / 120 / 100 / 80; T 7-14 use 200 / 140 / 100 / 80 / 60 / 40 exceptT 10 uses 120 / 80 / 60 / 40 / 20 / 10. T 7 weaker age 1 first visibleT 8. Designed patterns do not establish causal onboarding effect or exact calendar topline masking month.
Order status	Ordinary price 800/variable 560/CAC 200. T 10 first-order price 640/CAC 250; 1250 attempts=1000 retained+150 RTO+100 pre-dispatch cancel. RTO 100 cost/zero revenue; cancel zero cost/zero revenue. CAC excludes modeled fulfillment/RTO costs. No taxes/fixed/other adjustments/time discounting.
Horizon economics	Baseline age 3 contribution 379200/net after CAC 179200; T 10 initial 65000/age 3 total 127400/net-122600. These observed-horizon contribution sums are not lifetime value, cash flows, profit or guaranteed payback. Baseline clears CAC at age 0; T 10 not by age 3, future unknown.
Read then transfer	Use Inputs/Patterns -> Triangle/Rates and Bridge/Horizon. Static notes describe restored baseline only. Actual plan blank. Forecasts, GMVdefinitions, recognition, rights/audience/accepted owner require separate evidence; no mandatory dashboard access or real data upload.

Inputs | Observation and separate economics

Yellow inputs editable. Whole cohort sizes/counts. As-of 1-14; future remains unknown. Rupees are stipulated classroom amounts.

Observation end, calendar T number	14.00
Distinct initial retained buyers per cohort	1000.00

Ordinary retained order price rupees	800.00
Retained order variable cost rupees	560.00
Ordinary CAC per acquired buyer rupees	200.00
T10 retained initial order price rupees	640.00
T10 CAC per acquired buyer rupees	250.00
T10 RTO attempts at acquisition	150.00
Cost per T10 RTO rupees	100.00
T10 cancelled before dispatch attempts	100.00

Observation numeric gate	1.000000
Economics numeric gate	1.000000

Zero buyers leaves rates/per-buyer values undefined. Initial buyers are successful retained orders, not all checkout attempts. An economic loss is valid arithmetic, not invalid input. Numeric gates do not validate identity, status, causality, accounting or authority.

Patterns | Monthly distinct repeat buyers

Each cell is one monthly count, not a probability or disjoint set of people. Counts whole/nonnegative and no greater than cohort size.

Pattern / cohort	Age1	Age2	Age3	Age4	Age5	Age6
T1-T6 baseline	250.00	180.00	150.00	120.00	100.00	80.00
T7-T14 except T10	200.00	140.00	100.00	80.00	60.00	40.00
T10 special	120.00	80.00	60.00	40.00	20.00	10.00

Pattern numeric gate	1.000000

No person IDs supplied. Sum of monthly counts cannot establish unique-ever-repeat buyers, time-to-first-repeat, KM survival, subscription churn or population uncertainty. No age 7+ values invented. Unchanged older-cohort patterns are not current company evidence.

Triangle | Equal acquisition ages

N/A means interval not reached as of cutoff, not zero. Ages 0-6 only. No age 7+ or full lifetime supplied.

Cohort	Age0	Age1	Age2	Age3	Age4	Age5	Age6
T1	1000.000000	250.000000	180.000000	150.000000	120.000000	100.000000	80.000000
T2	1000.000000	250.000000	180.000000	150.000000	120.000000	100.000000	80.000000
T3	1000.000000	250.000000	180.000000	150.000000	120.000000	100.000000	80.000000
T4	1000.000000	250.000000	180.000000	150.000000	120.000000	100.000000	80.000000
T5	1000.000000	250.000000	180.000000	150.000000	120.000000	100.000000	80.000000
T6	1000.000000	250.000000	180.000000	150.000000	120.000000	100.000000	80.000000
T7	1000.000000	200.000000	140.000000	100.000000	80.000000	60.000000	40.000000
T8	1000.000000	200.000000	140.000000	100.000000	80.000000	60.000000	40.000000
T9	1000.000000	200.000000	140.000000	100.000000	80.000000	60.000000	N/A
T10	1000.000000	120.000000	80.000000	60.000000	40.000000	N/A	N/A
T11	1000.000000	200.000000	140.000000	100.000000	N/A	N/A	N/A
T12	1000.000000	200.000000	140.000000	N/A	N/A	N/A	N/A
T13	1000.000000	200.000000	N/A	N/A	N/A	N/A	N/A
T14	1000.000000	N/A	N/A	N/A	N/A	N/A	N/A

Age 0=successful retained first orders. Repeats one retained order per monthly buyer. T 7 age 1 first visibleT 8; T 10 age 6 not observed atT 14.
Static baseline notes must change if inputs change.

Rates | Exactly age 1, 3, 6

Fractions, not percent points. Same original buyer denominator; eligibility varies with age. Not on-or-after/ever-repeat/subscription survival.

Cohort	Age1 fraction	Age3 fraction	Age6 fraction
T1	0.250000	0.150000	0.080000
T2	0.250000	0.150000	0.080000
T3	0.250000	0.150000	0.080000
T4	0.250000	0.150000	0.080000
T5	0.250000	0.150000	0.080000
T6	0.250000	0.150000	0.080000
T7	0.200000	0.100000	0.040000
T8	0.200000	0.100000	0.040000
T9	0.200000	0.100000	N/A
T10	0.120000	0.060000	N/A
T11	0.200000	0.100000	N/A
T12	0.200000	N/A	N/A
T13	0.200000	N/A	N/A
T14	N/A	N/A	N/A
Eligible cohort count	13.000000	11.000000	8.000000

At baseline eligible 13/11/8. BaselineT 1=.25/.15/.08; T 7=.2/.1/.04; T 10 age 6 N/A. Empty or invalid observed denominators are errors, not zero-retention observations. Pooling unequal-age cells changes the question.

Bridge | T 10 acquisition status

Rupees. Stipulated status treatment only, not current statutory recognition policy or a GMV fraud finding.

Retained first-order buyers / orders	1000.000000
RTO attempts	150.000000
Pre-dispatch cancellations	100.000000
Total acquisition attempts	1250.000000

Retained initial net sales rupees	640000.000000
Retained variable cost rupees	560000.000000
RTO cost rupees	15000.000000
Initial contribution rupees	65000.000000
Acquisition cost rupees	250000.000000
Initial net after CAC rupees	-185000.000000

Baseline 1250 attempts=1000 retained+150 RTO+100 cancel; 640000 sales-560000 retained cost-15000 RTO=65000 contribution; after 250000 CAC=-185000. Failed attempts produce zero sales in this toy. No taxes, fixed costs, other adjustments, cash timing or real refund review.

Horizon | Cumulative through age 3

Baseline T 1 versus T 10, only observed intervals. Rupees and rupees per original retained buyer. N/A if cutoff too early, not zero.

Measure	T1 baseline	T10 special
Age0 contribution	240000.000000	65000.000000
Through age1 contribution	300000.000000	93800.000000
Through age2 contribution	343200.000000	113000.000000
Through age3 contribution	379200.000000	127400.000000
Acquisition cost	200000.000000	250000.000000
Age3 net after CAC	179200.000000	-122600.000000
Age3 contribution per original buyer	379.200000	127.400000
Age3 net sales per original buyer	1264.000000	848.000000
Age3 cumulative / initial net sales ratio	1.580000	1.325000

Baseline 379200 contribution/179200 after CAC/379.2 perbuyer/1264 salesperbuyer/1.58 ratio. T 10=127400/-122600/127.4/848/1.325. Ratios not SaaS NRR/repeat people; not full LTV, cash flows or profit. Baseline clearsCACage 0; T 10 does not through age 3, future unknown. Static notes restored baseline only.

Interpretation | Observation is not lifetime

A one-page scoped diligence paragraph plus transfer prompts. No actual analyst conclusion, channel decision, customer contact or audit.

What the toy shows	Equal age comparisons distinguish designed weaker monthly repeat patterns from incomplete observation. T 7 age 1 visibleT 8, not acquisitionT 7. An exact calendar month where topline masks decay is unestablished; no age 7+ or calendar revenue series was supplied.
What the counts do not show	Repeat monthly people overlap across months.250+180+150=580 order events, not 580 unique repeaters. With no customer-event histories, cannot derive time-to-second-order, KM censoring risksets, subscription survival, causal effects or statistical significance.
Source wording conflict	Amplitude opening says Return On or After can measure whether users came back by a point, but its displayed numerator is on day X or any later day, not by X. Follow explicit event definition. This workbook is exactly-age monthly repeat activity, not that on-or-after cumulative-tail metric.
Economic scope	T 10 promotional initial price plus separately charged RTO lowers initial contribution; ordinary repeat price remains 800. After CAC net remains negative atage 3. CAC payback is not total business break-even/profit; positive contribution does not settle fixed costs, tax, cash timing, refunds or future behavior.
Closed-note transfer	Change observation end toT 12: eligibleage 1/3/6=11/9/6; T 10 age 3 N/A. RestoreT 14, then raiseRTO cost to 200: initial 50000/age 3 contribution 112400/net-137600. Explain why rate cells do not change from anRTO cost-only edit. Actual learner answer blank.
Next evidence	Define identity, status/date/timezone, cohort start, qualifying event, age clock, horizon, refund/GMV/sales/cost basis and permitted audience. Same-age improvement is association; mix/seasonality/selection/price may differ. No guaranteed campaign lift or mandatory dashboard invitation.

Actual plan | Unknown remains blank

No real customer IDs, data extracts, source grants, accepted reviewer, learner result or accounting conclusion invented.

Definition / clock	Artifact / cutoff	Identity / status	Cost / sales basis	Contradiction	Accepted owner	Authority / next

Blank is unknown, not zero, pass, consent or completed diligence. No actual brand records, private upload, analytics invitation, account creation, customer contact, accounting clearance, marketing campaign or spend.

Evidence | Seven URLs, selected source scope

Six organizations/three types: practitioner/vendor guidance, government methods, accounting textbook. Written returned text, not full source AV or underlying studies.

Original framework	https://www.goingvc.com/post/a-guide-to-conducting-a-cohort-analysis-as-a-vc | Complete returned May 15, 2025 written article read, not images/linked resources. VC vintage/stage/sector and like-age/definition guidance adapted to explicit D 2 C toy, not fund outcomes or causal validation. Course price/program claims not used or independently validated.
Investor views	https://amplitude.com/blog/startup-fundraising-metrics | Complete returned February 21, 2023 written read, not images/cited outcomes/current product tiers. Investor quotations are stage/model-dependent views, not universal retention threshold or PMF proof. Dashboard sharing optional, not consent or diligence entitlement.
Official calculation	https://amplitude.com/docs/analytics/charts/retention-analysis/retention-analysis-calculation | Complete returned written read, not images/linked Academy. Return On exact day differs from on-or-after tail and incomplete eligible denominators. Opening by-point wording conflicts with displayed on-or-after formula; not silently treated as within-by X. No account used.
Applied methods	https://engineeringblog.yelp.com/2020/02/accelerating-retention-experiments-with-partially-observed-data.html | Complete returned February 20, 2020 written read, not images/referenced papers/simulations replicated. Reported 12-16%speedup contextual, not universal. https://ecomtoolkit.net/blog/ecommerce-repeat-purchase-right-censoring-cohort-analysis-2026/ | Complete returned September 9, 2026 written read, not images/linked resources. Illustrative KM not fit to our monthly counts; no service contact.
Methods and accounting	https://www.itl.nist.gov/div898/handbook/apr/section1/apr131.htm | Complete returned written censoring page read, not diagram/linked estimators. Reliabilitycontext adapted conceptually, not retail dropout model. https://openstax.org/books/principles-managerial-accounting/pages/3-key-terms | Complete returned key terms read, not whole chapter/book/statutory recognition review.
Limits	All case figures fictional. No actual cohort/investor/company/accounting/causal validation, current prices, private data, full source/AV, native Excel/Sheets parity, independent learner/account/field/actual 10. Original objectives/exercise/quiz retained. Actual evidence/authority/accepted owner blank.

Failure log | Authored checks only

Independent expected arithmetic, eligibility and controlled failure tests. Not actual learner mistakes, company outcomes or accounting clearance.

1	PASS: 174 baseline formulas independently recomputed.
2	PASS: 16 changed input sets plus restored baseline.
3	PASS: 20 invalid inputs/3480 isolation checks.
4	PASS: 6 broken references/6 checks.
5	PASS: 8 boundaries;zero denominators undefined.
6	PASS: future intervals N/A;no invented zero or age7+.
7	PASS: loss-making contribution retained;families isolated.
8	PASS: actual evidence/consent/accepted-owner rows blank.